[의약정보학]

Evidence Based Pharmacotherapy

Kyungim Kim, Ph.D., MHPE
College of Pharmacy, Korea University

I. Learning objectives

II. What is the EBM?

III. The PICO model

IV. Levels of evidence

V. The literature search

I. Learning objectives

I. Learning objectives

Students Will Be Able To

1.1 Define EBM/EBP

1.2 Express key differences between study types

1.3 Utilize the evidence pyramid and PICO

1.4 Know major types of drug information references

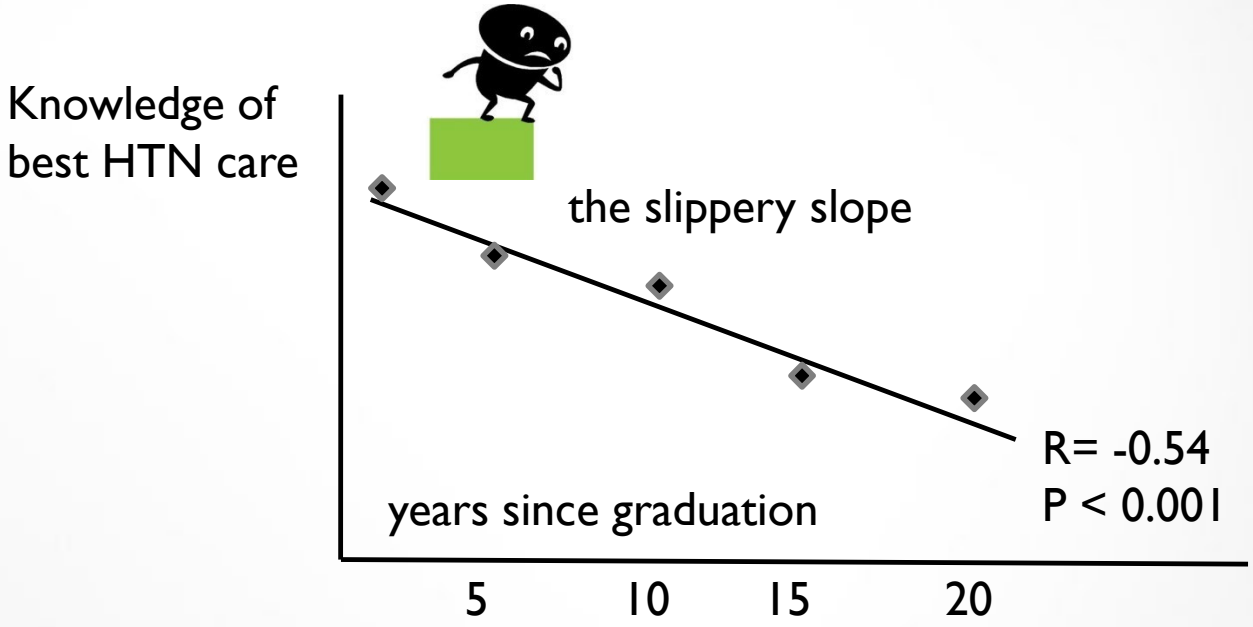
II. What is the EBM?

2.1 What is EBM/P?

- **The need for ( )**

  a. "Only about 15% of medical interventions are supported by solid scientific evidence!" [BMJ editorial, 1995]

  b. Evidence to stay up to date

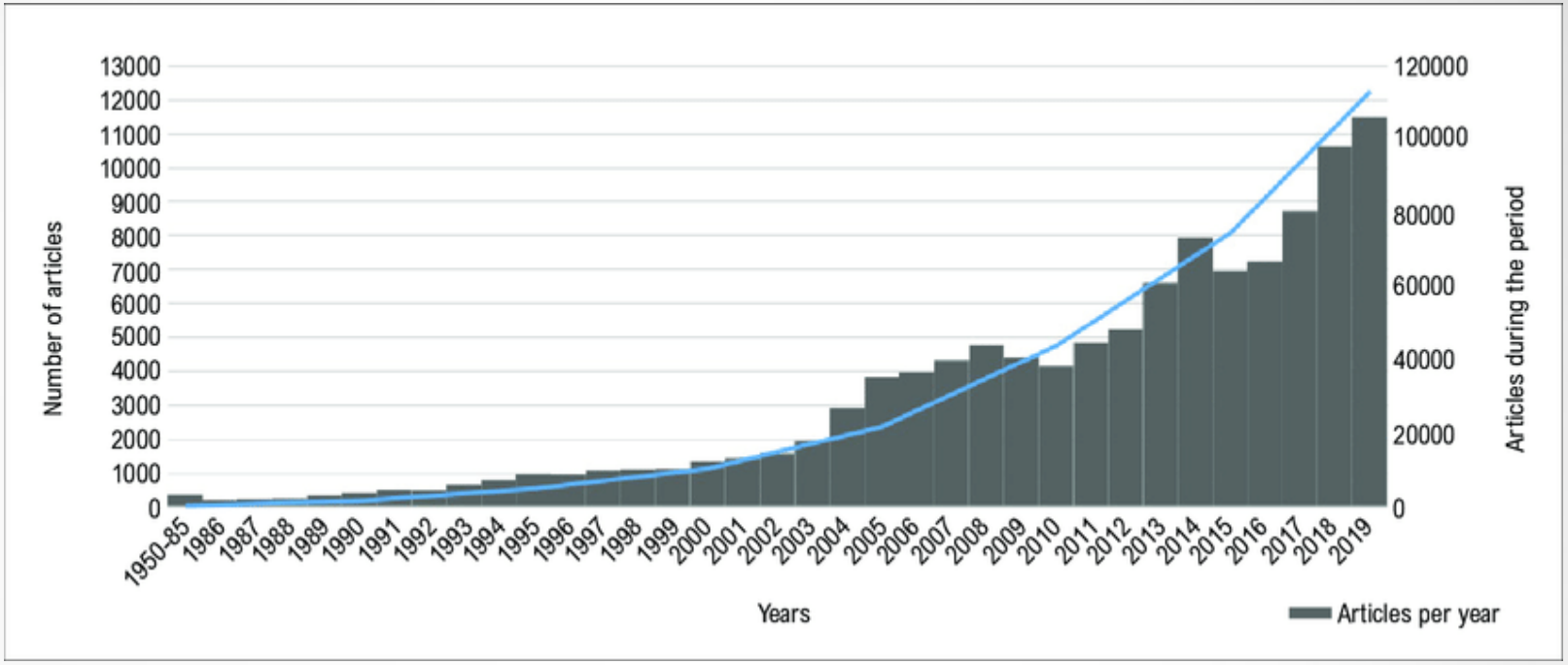


[자료 출처] Shiri. et al, (1993)

2.1 What is EBM/P?

- **The need for evidence**

  c. Besieged with information



[그림 출처] Manoel Horta Ribeiro, et al. (2022)

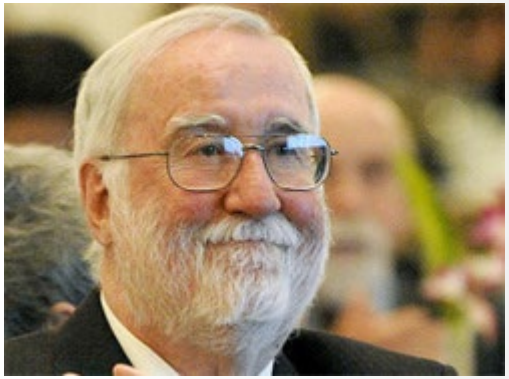
2.1 What is EBM/P?

- **The appearance of EBM**

  a. Proposed by Dr. David Sackett and colleagues at McMasters University in Canada

     - "The conscientious, explicit and judicious use of ( ) evidence in making ( ) about the care of the ( ) patient. It means integrating individual clinical expertise with the best available external clinical evidence from systematic research." (Sackett D, 1996)

  b. An important change in the way practice, teach, and do research

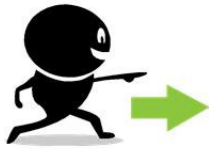


2.1 What is EBM/P?

- **The appearance of EBM**

Traditional paradigm

- ( ) experience
  - ✓Anecdote
  - ✓Apprenticeship
  - ✓Expert ( )
- Pathophysiologic rationale
- Clinical institution
- Possible benefit



EBP paradigm

- Knowledge of
  - ✓Strength of evidence
- ( ) statement of benefit and risk
- Patient's ( ) and preferences
- Informed decision-making

2.1 What is EBM/P?

- **Objectives of EBM**

a. Saving lives

- Apply ( ), filtered information to make informed decisions

b. Saving times

- So much information, too little time

c. Keeping your skills ( )

- Memory decrease
- New treatment methods

d. Supplementing clinical judgment

- Decisions should not be based only on institution, opinion or anecdotal information.

2.1 What is EBM/P?

- **EBM is the integration of**

a. Best ( )

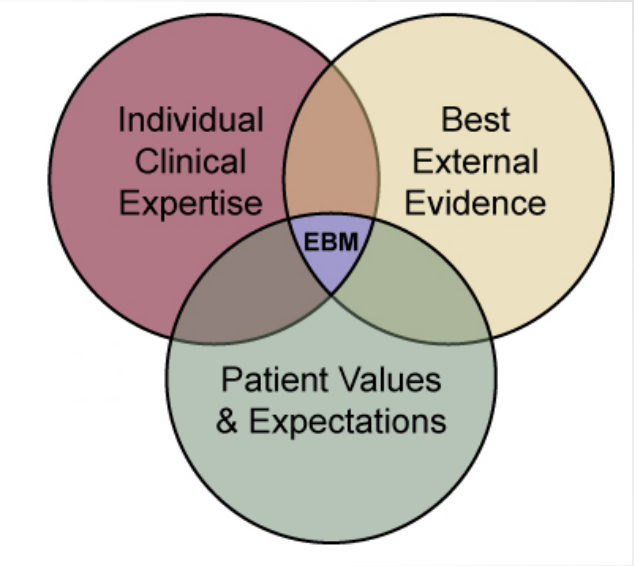
- Clinical relevant
- Patient oriented

b. Clinical ( )

- Use of clinical skills and past experiences to identify a patient's unique health status and needs

c. Patient ( )

- Unique preferences, concerns, & expectations each patient brings



2.1 What is EBM/P?

- **What EBM is NOT!!**
  a. Not "( ) medicine"
  b. External evidence can inform, but never replace individual clinical expertise
  c. EBM is not cost-cutting medicine
  d. EBM is not restricted to randomized trials and meta-analysis



2.1 What is EBM/P?

- **EBM – clinical strategy**

  a. Evidence-based health care (EBHC)

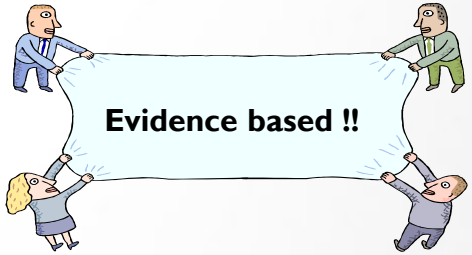
  - Extends the application of the principles of EBM to all professions associated with health care (CEBM, Oxford, 2004)

  b. Adaptation across several practice areas, fields, disciplines

  - Evidence-based pharmacotherapy

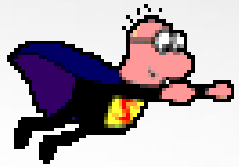
    "an approach to decision making whereby clinicians appraise the scientific evidence and its strength in support of their therapeutic decisions"

  - Nursing, public health, clinical practice, etc.



2.2 Steps of EBM/P

- **Steps in the EBP process**



( ) your patients

( ) clinical questions

( ) the best evidence

Appraise the evidence

Apply evidence to patient care

( )

2.2 Five steps of EBM/P

- **Five steps of EBM** [Sackett DL et al, 2000]

Five steps of EBM	
Ask the question	**Need to information about a patient problem and formulate a clear, concise, clinical question (3C's)**
Acquire the evidence	**Search the literature for best evidence to help you answer the question**
Appraise the evidence	**Critically appraise the evidence for its validity and applicability**
Apply the evidence	**Integrate this critical appraisal with clinical expertise and patient information**
Evaluation	**Evaluate the effectiveness and efficiency in executing steps 1-4 and seek ways to improve them both for the next time**

III. The PICO model

3.1 The well-built clinical question

- **( ) in the EBM process**
- **( ) of a good clinical question: ( )**

a. A mnemonic that helps one remember the key components of a well focused question

b. The question needs to identify the ( ) of the patient, what treatment you are considering for the patient, what alternative treatment is being considered (if any) and what is the outcome you want to avoid or promote.

- **The most common types of question**

a. Diagnosis

b. Therapy

c. Prognosis

d. Harm/etiology



Case #1

- This is a new patient who recently moved to the area to be closer to her son and his family. CP is 73 years old and has a history of congestive heart failure and a left ventricular ejection fraction of 40%.
- She has been hospitalized twice within the last 9 months for worsening of heart failure. She is extremely diligent about taking her medications (enalapril, aspirin and simvastatin) and wants desperately to stay out of the hospital. She lives alone with several cats and a canary.
- She comes to you with an advertisement for Aprovel (Irbesartan) and wants to know if this would help her. You are not certain of the evidence supporting this drug. You decide to research this question before discussing this with her during the next visit.



3.2 Construct a question using the PICO model

PICO model	
P = ( ) or problem	• How would you describe a group of patients similar to yours? What are the most important characteristics of the patient? • This may include the primary problem, disease, or co-existing conditions. • Sometimes the sex, age or race of a patient might be relevant to the diagnosis or treatment of a disease.
I = ( ), prognostic factor, or exposure	• Which main intervention, prognostic factor, or exposure are you considering? What do you want to do for the patient? Prescribe a drug? Order a test? Order surgery? • What factor may influence the prognosis of the patient? Age? Co-existing problems? • Has the patient been exposed to something? Asbestos? Cigarette smoke?

3.2 Construct a question using the PICO model

PICO model	
C = ( )	• What is the main alternative to compare with the intervention? • Are you trying to decide between two drugs, a drug and no medication or placebo, or two diagnostic tests? • Your clinical question does not always need a specific comparison.
O = ( )	• What can you hope to accomplish, measure, improve or affect? • What are you trying to do for the patient? Relieve or eliminate the symptoms? Reduce the number of adverse events? Improve function or test scores?

Case #1

- For this patient, the clinical question might be

“ ?”



IV. Levels of evidence

4.1 The evidence pyramid

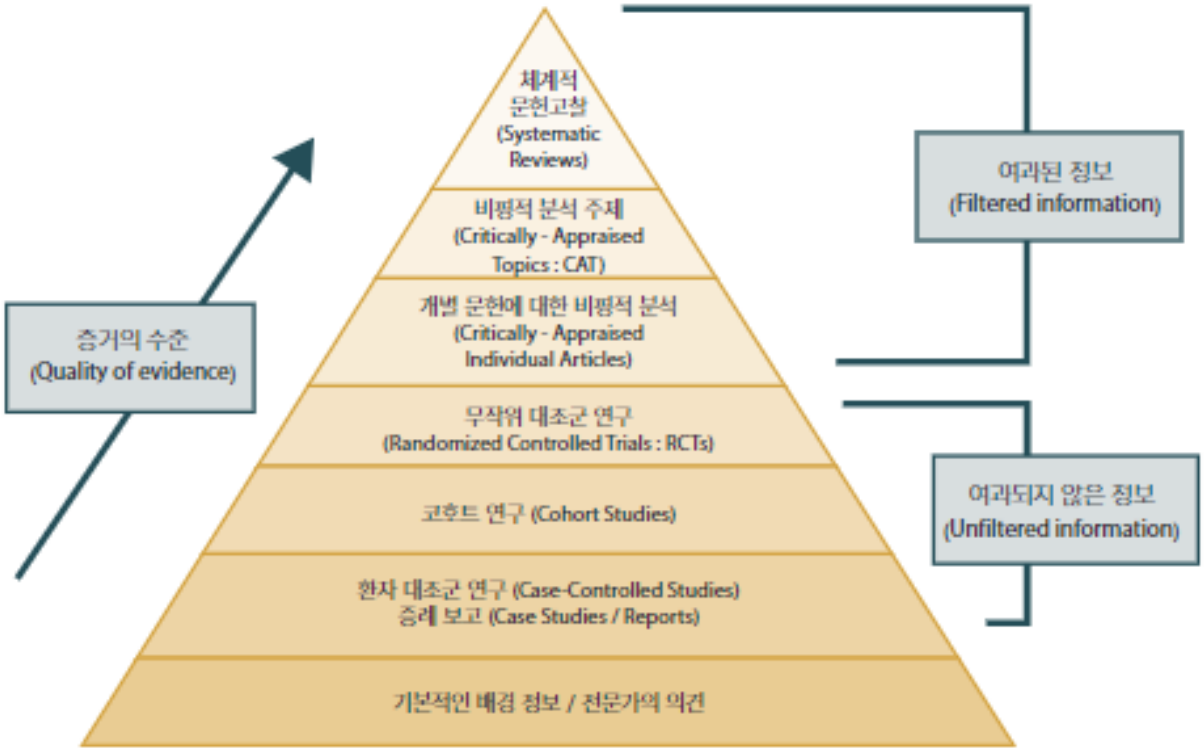
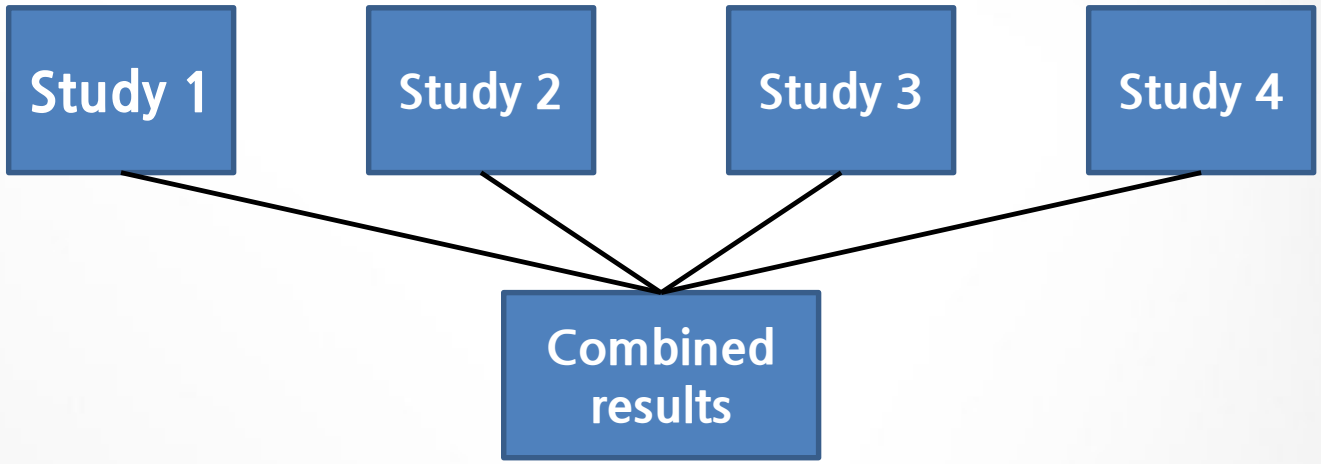


그림 27-1 근거 수준의 피라미드(Pyramid of Evidence)[4]

[그림 출처] 임상실무약학, 2024

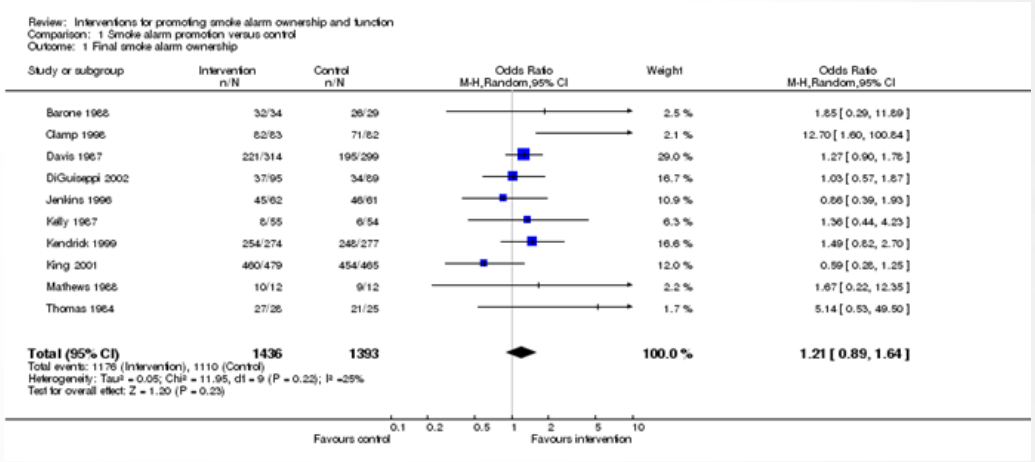
4.2 Study design

- ( )
  a. Descriptive summary of previous studies
  b. An expert has used a specific, ( ) of inclusion
     - E.g., all RCTs involving treatment of schizophrenic adults with Drug X that were conducted in psychiatric units



4.2 Study design

- **Meta-analysis**
  a. "Best evidence"
  b. The result of an exhaustive search of the medical literature, selecting only valid, randomized controlled trials pertaining to a specific topic.
  c. Data are combined to increase ( ) and increase clinicians' confidence in the conclusions using complex statistical methodology
     - E.g., combination of data from 6 case-control studies about smoke exposure and cancer incidence



[그림 출처] https://training.cochrane.org/handbook/current/chapter-10

4.2 Study design

- **Randomized controlled clinical trials**
  a. ( ), prospective studies
  b. Participants are ( ) assigned to experimental and control groups
  c. A method to prevent bias is called ( ).
  d. Good for treatment or prevention questions
     - drug A v. placebo
     - drug A v. drug B v. placebo
  e. ( )
     - E.g. MI, death
  f. Surrogate marker
     - E.g. LDL-C change %

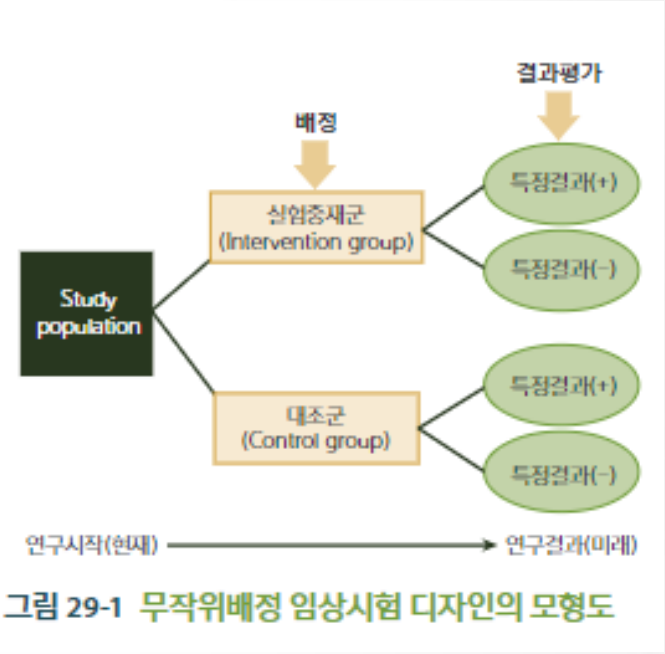


그림 29-1 무작위배정 임상시험 디자인의 모형도

[그림 출처] 임상실무약학, 2024

4.2 Study design

- **Cohort studies**

  a. Groups are ( ) to determine if exposure to something (i.e., smoking) or a health condition (i.e., obesity) will result in a certain effect. (study)

     - E.g., drug A v. no treatment over 10 year period

  b. Prospective cohort vs. Retrospective cohort

  c. Less reliable than RCT (conditions)

  d. Good for etiology questions or when RCT would be unethical

  e. Good when exposures can't be controlled or when outcomes occur infrequently

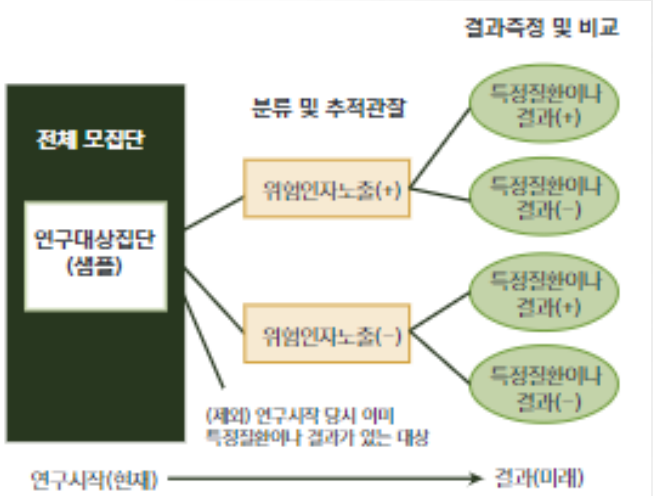


그림 29-2 전향적 코호트 연구 디자인의 모형도

[그림 출처] 임상실무약학, 2024

4.2 Study design

- **Case-control studies**

  a. Individuals with condition are compared to those without to determine if exposure to something caused the condition

     - E.g., men with prostate cancer v. those without

  b. ( ), ( ) study

  c. Difficult to determine
  how much exposure may have caused the condition

  d. Good for etiology questions,
  ( ) or
  when it may take a long time
  for the disease to develop

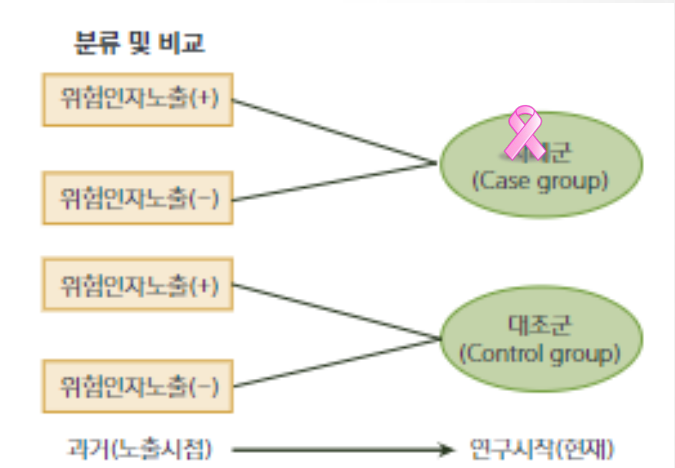
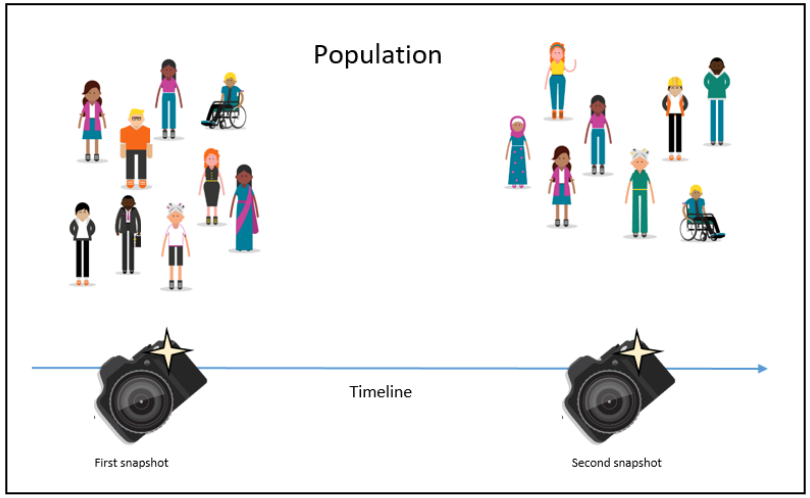


그림 29-3 환자-대조군 연구 디자인의 모형도

[그림 출처] 임상실무약학, 2024

4.2 Study design

- **( ) studies**
  a. Observational studies that analyze data from a population at a ( )
  b. Often used to measure the ( ) of health outcomes, understand determinants of health, and describe features of a population
  c. Unable to assess ( ), to study rare diseases, and to make a ( ) inference



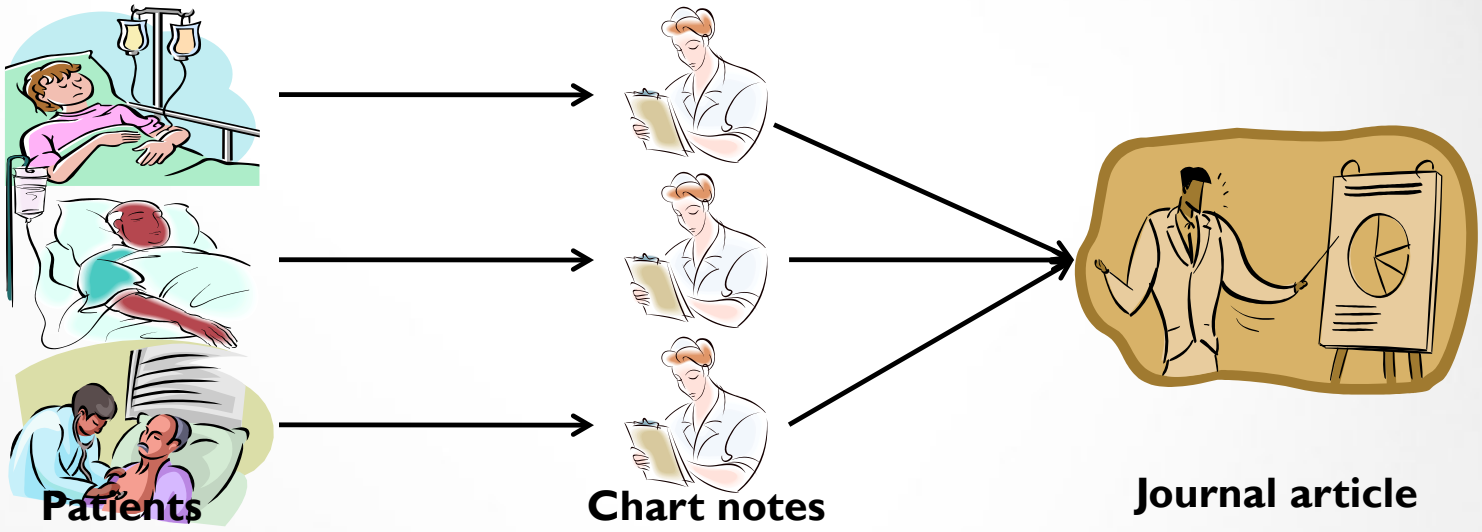
[그림 출처] https://deakin.libguides.com/quantitative-study-designs/cross-sectional

4.2 Study design

- **Case series**

  a. A descriptive study that tracks patients with a known exposure given similar treatment

  b. Although case series are considered very weak evidence, they serve the purpose of introducing the practitioner to rare, unusual conditions.



4.2 Study design

- **Editorial and expert opinions**

  a. Although editorials and expert opinions play a role in evidence-based practice, keep in mind that they are ( ), and therefore are not considered strong evidence.

  b. There is ( ), and therefore the results have no ( ) value.

  c. However, the experience of experts is valuable and may be necessary to make a clinical decision

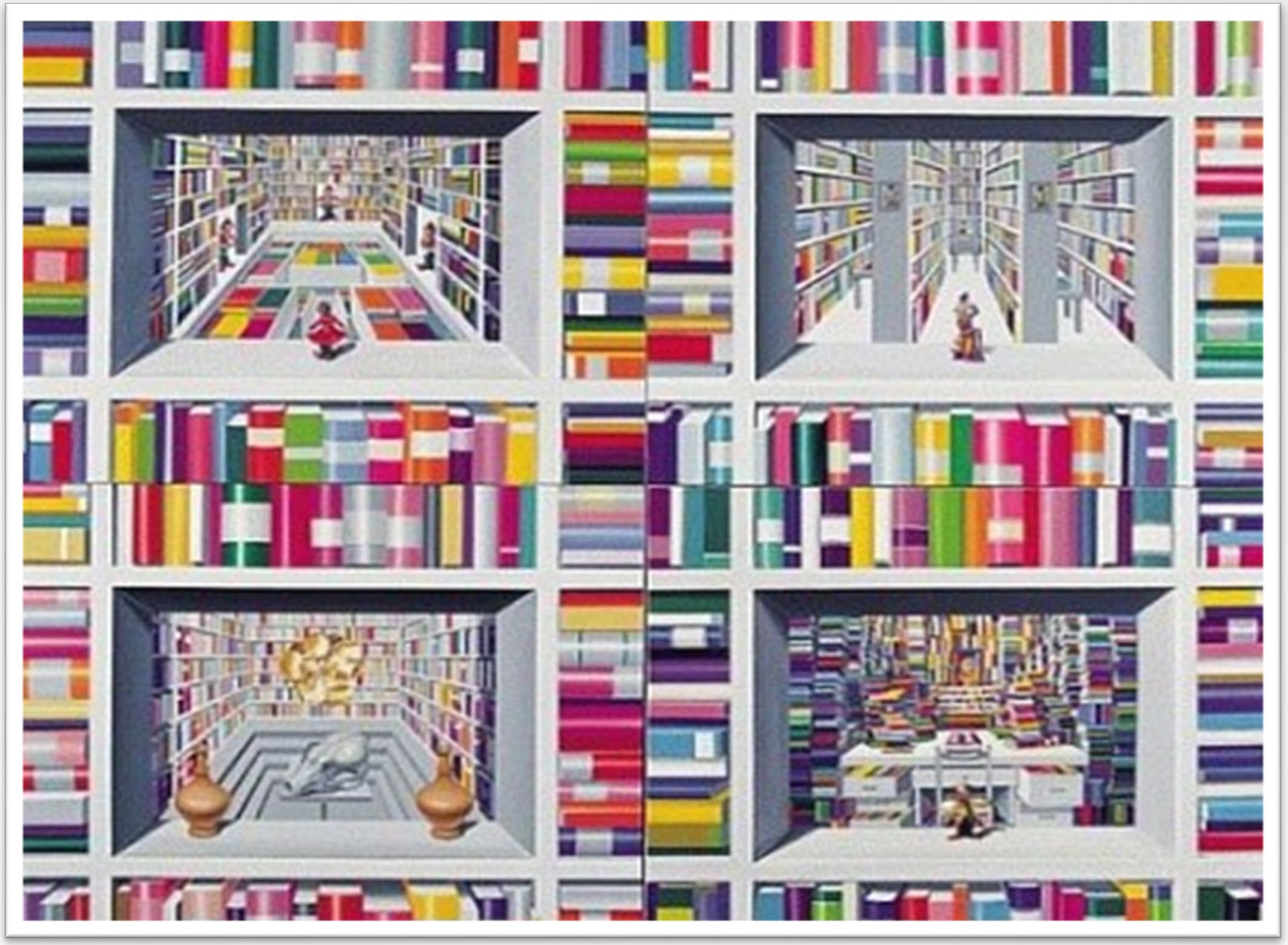
V. The literature search

5.1 Importance of literature searching skills



‘책 정물’ 라이덴의 화가, 1628년경

5.1 Importance of literature searching skills



‘서재4’ 홍경택, 2004

5.1 Importance of literature searching skills

- **Pharmacists as a drug informatics specialists**
  a. 환자를 늘 접하는 약사로서 의약품에 관련된 정보의 확보와 적절한 활용은 환자의 진료 및 치료에 중요함
  b. 약사직능의 경쟁력은 약사들의 의약정보 관리능력에 좌우됨
  c. 의약분업과 급변하는 의료환경은 약사들의 의약정보제공 및 관리능력을 요구함

5.1 Importance of literature searching skills

- **What is the ideal resource?**
  a. Located in the clinical setting
  b. Easy to use
  c. Fast, reliable connection
  d. Comprehensive /Full Text
  e. Effective search engine
  f. Provides primary data

5.2 Types of information references

- **1차 문헌**

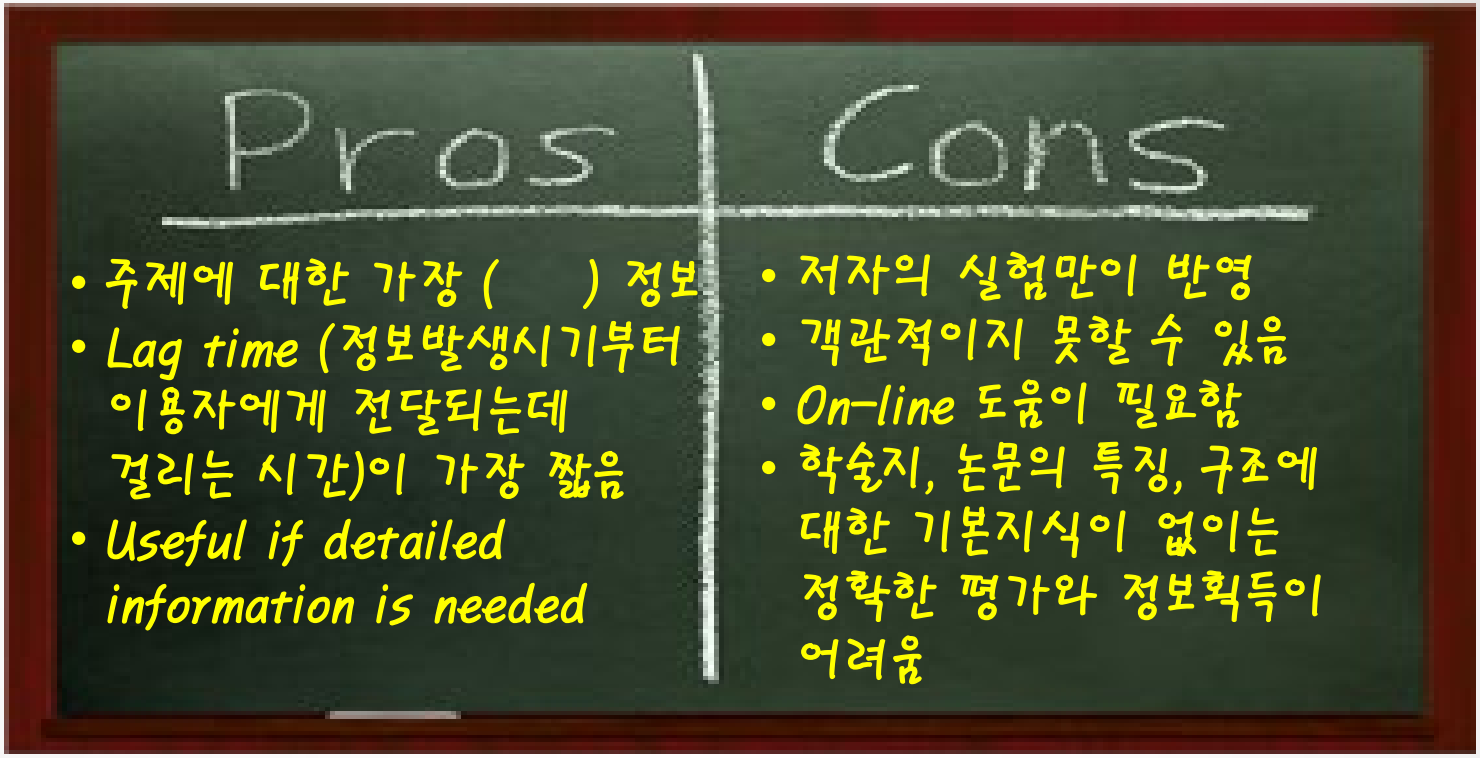
a. 1차 문헌의 특징

- 정기간행물 (Journal), 최신정보
- 기존의 지식을 늘리거나 새로운 지식을 소개하기 위해 발표되는 ( )
  : original research, case reports, etc.
- 2차, 3차 문헌을 구성하는 기초자원
- 최신 정보를 담고 있으므로 전문성을 유지하기 위해서는 지속적 접근이 필요
  : e-mail alerts for table of contents

5.2 Types of information references

- **1차 문헌**

  b. 1차 문헌의 장점 및 단점



5.2 Types of information references

- **1차 문헌**

  c. 1차 문헌의 평가

  - Journal name과 저자(발표자)에 의해 출판물을 확인
  - Journal의 목적
  - 논문의 종류

    : original research, case reports, comments, letter, etc.
  - Peer review
  - Publication frequency

5.2 Types of information references

- **1차 문헌**

d. Peer-reviewed pharmacy journals

표 28-1 1차 문헌

학술지명	논문분류	논문내용	발행빈도	인용 정보원	출판사
American Journal of Hospital Pharmacy (AJHP)	a,b,c,d,e,f,g	1,2,3,4,5, 6,7,8,9,10, 11,12,13	격월	a,c,d,e,f,g	American Society of Health-System Pharmacists
Annals of Internal Medicine	a,b,c,d,e,f,g	2,3,6,9, 11,12,13	격월	a,f,g	American College of Physicians
Annals of Pharmacotherapy	a,b,c,d,e,f,g	1,2,3,4,5, 8,9,10,11	월간	a,b,c,e,f,g	Harvey Whitney Books Company
British Medical Journal (BMJ)	a,b,c,d,e,f,g	4,6,7,9,11	주간	c,d,f,g	British Medical Association House
Drug Information Journal	b,d,e,g	4,6,11,12	계간	a,d,e,f	Drug Information Journal
Hospital Formulary	a,b,c,d,e,f	2,3,5	월간	a,b,c,d,e,f	Advanstar Communications Inc.
Hospital Pharmacy-Lippincott's	a,b,c,d,e,f	1,2,3,7,8,10, 11,12,13,14	월간	a,b,c,d,e,f,	J.B.Lippincott Company
Journal of Parenteral and Enteral Nutrition (JPEN)	b,c,d,e,f	3,6,9,10,11	격월	d,e,g	American Society for Parenteral and Enteral Nutrition
Journal of Pediatrics	a,b,c,d,e,f,g	3,4,9,10,11	월간	a,b,c,d,e,f,g	Mosby
Journal of the American Medical Association (JAMA)	a,b,c,d,e,f,g	4,6,7,9,10,11	주간	a,b,c,d,e,f,g	American Medical Association
Lancet	a,b,d,e,f	2,3,4,9,10,11	주간	a,b,c,d,e,f,g	Williams & Wilkins
New England Journal of Medicine	b,c,d,e,f,g	2,3,4,6,9,11	주간	a,b,c,d,e,g	Massachusetts Medical Society

[그림 출처] 임상실무약학, 2024

5.2 Types of information references

- **2차 문헌**

  a. 2차 문헌의 특징

  - 1차 문헌의 ( )를 수집하여 데이터베이스로 정리해 놓은 검색정보원
    : Available in print, CD-ROM, or online DB (PubMed, MEDLINE)
  - 검색을 통해 1차 문헌으로의 직접적 접근을 허용함
    : 제목, 저자, 발행연도, 저널 이름, 주요어 등으로 검색이 가능

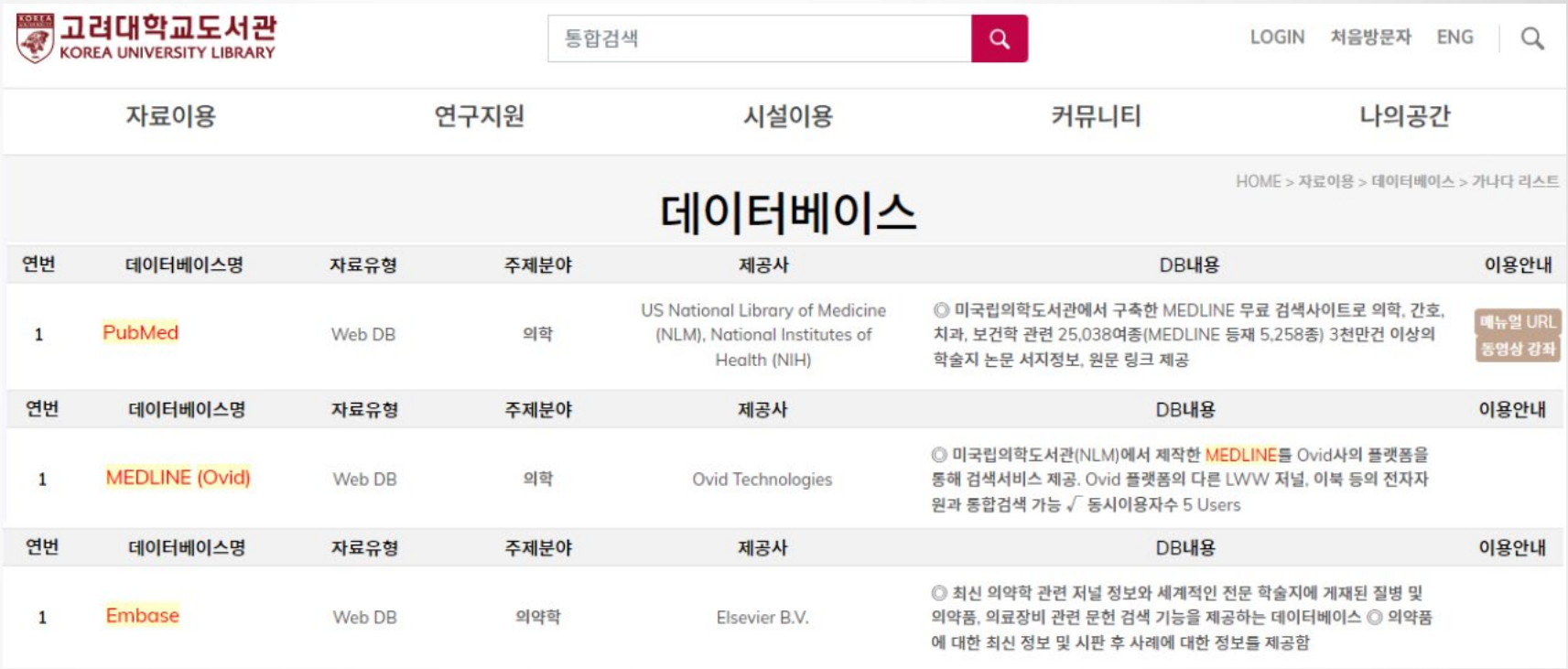
표 28-2 2차 문헌

Database	형태	종류	발행빈도	Unique Features	출판사
Biosis (Biosciences Information Service) (PV: Biological abstracts)	1,2,3	a,c	주간	a,b,c,d,e	BIOSIS
Cancerlit	1,3	a,c	월간	a,b,c,d,e	National Library of Medicine
CA Search (PV: Chemical Abstracts)	1,2,3	a,c	격주간	a,b,d	Chemical Abstract Services
Current Contents: Life Sciences	1,2	a,c	주간	b,d,e	Clarivate Analytics
EMBASE (PV: Excerpta Medica)	1,2	a,c,d	주간	b,d,e	Elsevier Science, Inc.
IPA (PV: International Pharmaceutical Abstracts)	1,2,3	a,b	격주간	a,b,c,d,e	Clarivate Analytics
MEDLARS Medline (PV: Index Medicus)	1,2,3	a,c	주간	a,b,c,d,e	National Library of Medicine

[그림 출처] 임상실무약학, 2024

5.2 Types of information references

- **2차 문헌**
  a. 2차 문헌의 특징



고려대학교도서관 KOREA UNIVERSITY LIBRARY

통합검색

LOGIN 처음방문자 ENG

자료이용 연구지원 시설이용 커뮤니티 나의공간

HOME > 자료이용 > 데이터베이스 > 가나다 리스트

데이터베이스

연번	데이터베이스명	자료유형	주제분야	제공사	DB내용	이용안내
1	PubMed	Web DB	의학	US National Library of Medicine (NLM), National Institutes of Health (NIH)	◎ 미국립의학도서관에서 구축한 MEDLINE 무료 검색사이트로 의학, 간호, 치과, 보건학 관련 25,038여종(MEDLINE 등재 5,258종) 3천만건 이상의 학술지 논문 서지정보, 원문 링크 제공	메뉴얼 URL 동영상 강좌

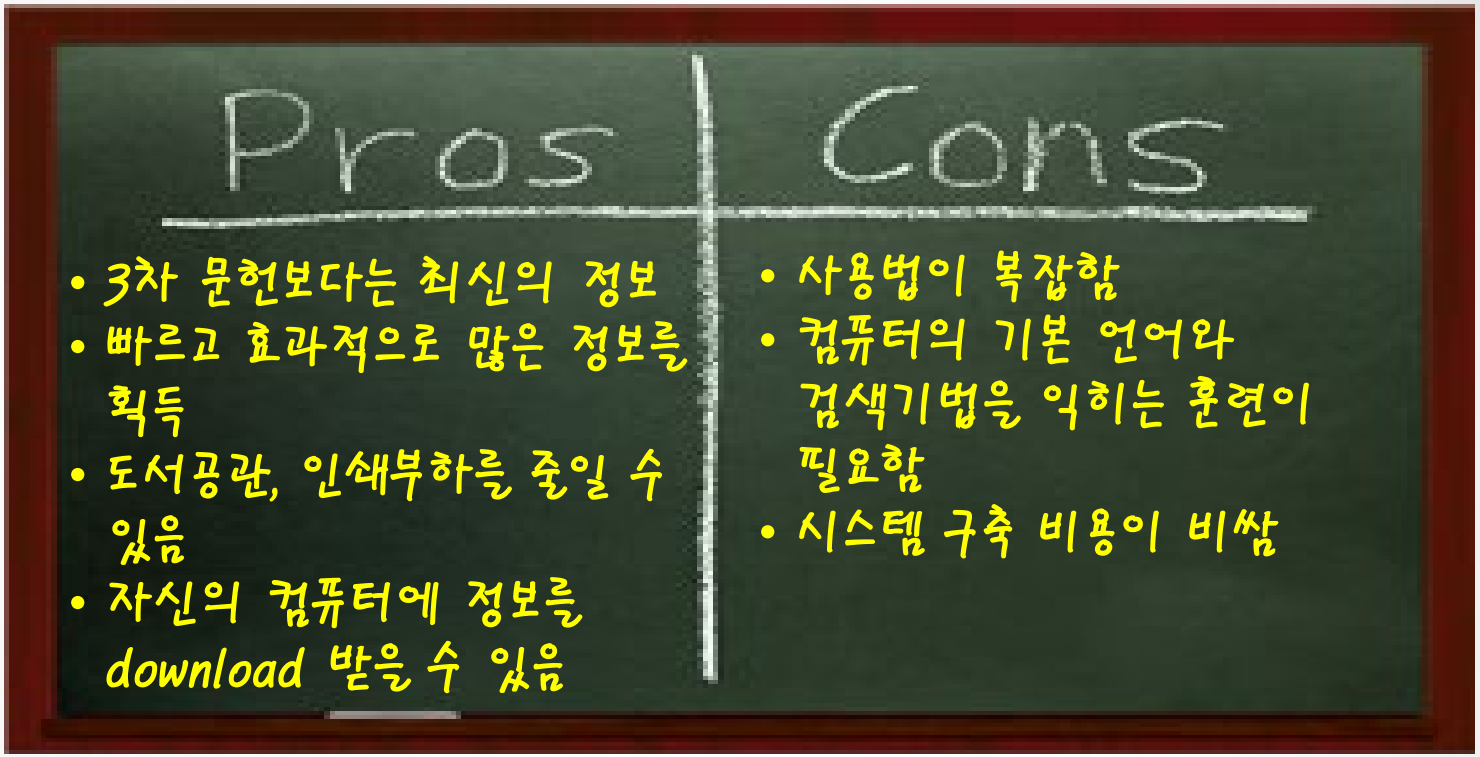
연번	데이터베이스명	자료유형	주제분야	제공사	DB내용	이용안내
1	MEDLINE (Ovid)	Web DB	의학	Ovid Technologies	◎ 미국립의학도서관(NLM)에서 제작한 MEDLINE를 Ovid사의 플랫폼을 통해 검색서비스 제공. Ovid 플랫폼의 다른 LWW 저널, 이북 등의 전자자원과 통합검색 가능 √ 동시이용자수 5 Users	

연번	데이터베이스명	자료유형	주제분야	제공사	DB내용	이용안내
1	Embase	Web DB	의약학	Elsevier B.V.	◎ 최신 의약학 관련 저널 정보와 세계적인 전문 학술지에 게재된 질병 및 의약품, 의료장비 관련 문헌 검색 기능을 제공하는 데이터베이스 ◎ 의약품에 대한 최신 정보 및 시판 후 사례에 대한 정보를 제공함	

5.2 Types of information references

- **2차 문헌**

  b. 2차 문헌의 장점 및 단점



5.2 Types of information references

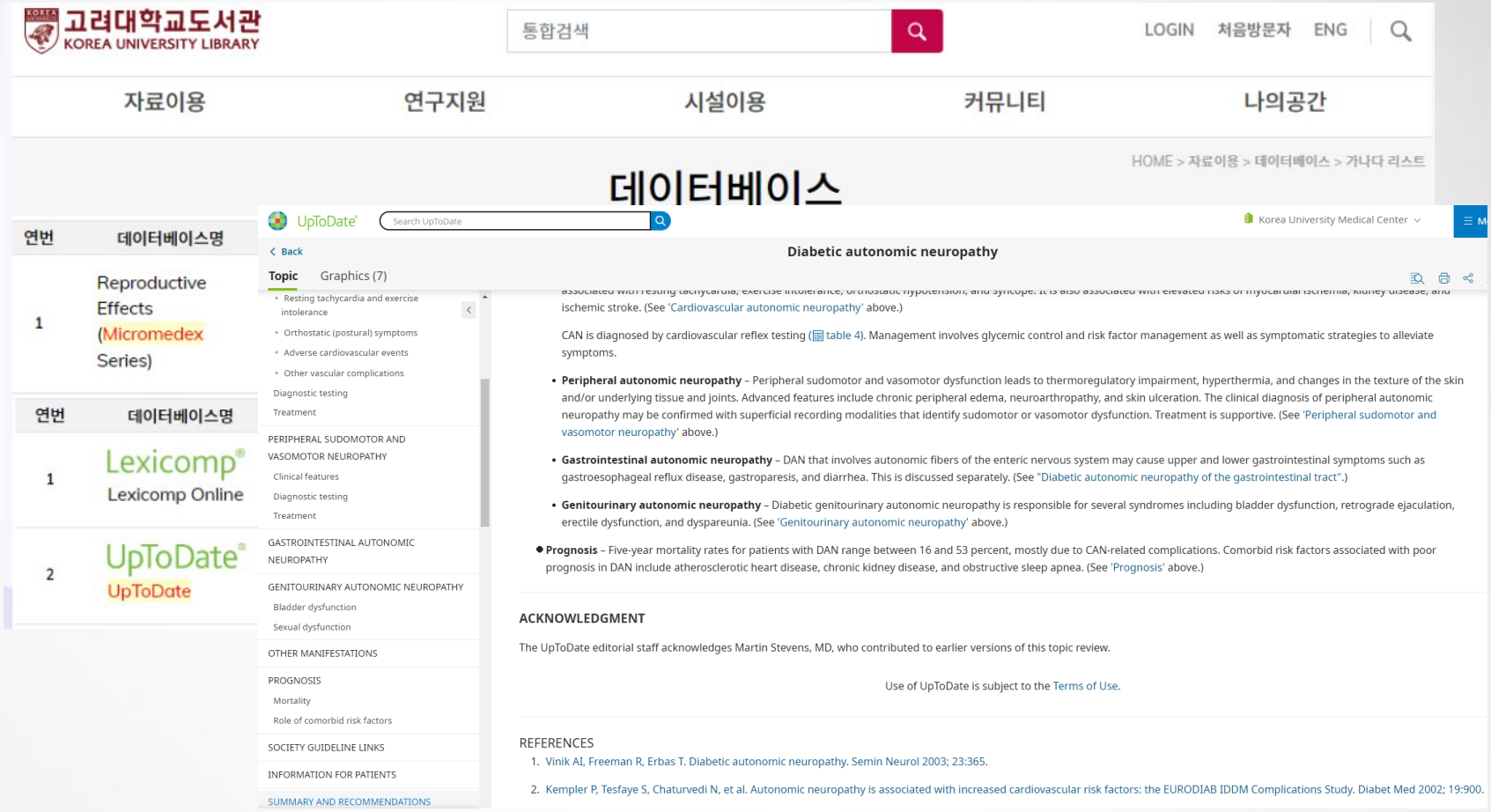
- **3차 문헌**

a. 3차 문헌의 특징

- 1차 및 2차 문헌 등 다양한 정보를 수집, 정리, 평가하여 종합한 자료
  - : ( ), 사전류, 의약품집, 종설논문 (review article)
  - : 전문데이터베이스 (Micromedex, UpToDate, Lexicomp)
- Available in many forms
  - : print, CD-ROM, or online DB

5.2 Types of information references

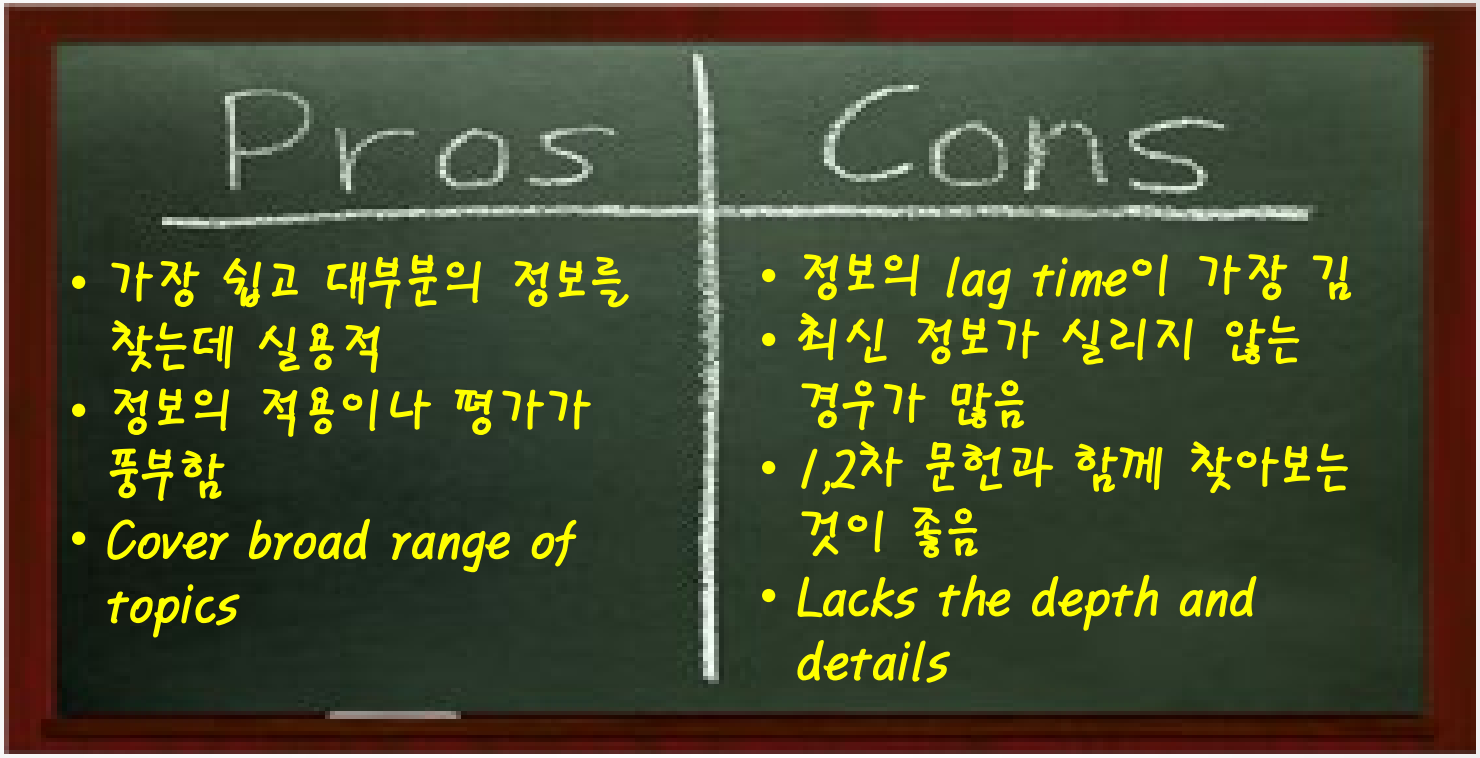
- **3차 문헌**
  a. 3차 문헌의 특징



5.2 Types of information references

- **3차 문헌**

  b. 3차 문헌의 장점 및 단점



5.2 Types of information references

- **3차 문헌**
  - c. 3차 문헌의 종류

표 28-3 3차 문헌

서명	질문 분류	형태	발행
AHFS DI (American Hospital Formulary Service Drug Information)	a,c,e,f,g,h,i,j,l,o,p,q,r,s	3	American Society of Health-System Pharmacists
Applied Therapeutics: The Clinical Use of Drugs	a,e,f,g,h,i,j,l,o,p,r,s	3	Applied Therapeutics, Inc.
Basic Skills in Interpreting Laboratory Data	a,f,g,h,i,o,s	3	American Society of Health-System Pharmacists
Clinical Toxicology of Commercial Products: Acute Poisoning	a,b,k,q,s	3	Williams & Wilkins
Drugdex	a,b,c,d,e,f,g,h,i,j,k,m,n,o,p,q,r,s	2,3	Micromedex, Inc.
Drug Evaluation (American Medical Association)	a,e,f,g,h,i,j, l,m,o,p,q,r	3	W.B. Saunders Company
Drug Facts and Comparisons	a,b,e,f,i,j,k,l,m,o,p,r,s	2,3	Facts and Comparisons
Drugs in Pregnancy and Lactation: A Reference Guide to Fetal and Neonatal Risk	a,g,q,s	3	Williams & Wilkins
Drug Interaction Facts	a,e,f,g,h,i,l,o, p,r,s	3	Facts and Comparisons
Goodman and Gilman's Pharmacological Basis of Therapeutics	a,j,n,o,p,q,r,s	3	McGraw-Hill, Inc.
Handbook on Injectable Drugs	c,e,f,l,m	1,3	American Society of Health-System Pharmacists
Martindale's: The Extra Pharmacopoeia	a,b,d,e,f,g,h,i,j,k,m,n,o,p,q,r,s	1,3	Pharmaceutical Press
Merck Index	b,c,d,k,m,n,o,p,s	3	Royal Society of Chemistry
Physician's Desk Reference (PDR)	a,b,c,d,e,f,g,h,l,k,l,o, p,q,r,s	1,3	Medical Economics, Inc.
Poisindex	a,e,k,l,m,n,r,s	2,3	Micromedex, Inc.
Remington's Pharmaceutical Sciences	a,c,d,e,j,k,p,r,s	3	Mack Publishing Company
Drug Information Handbook	a,b,d,e,f,g,h,i,l,o,p,q,r,s,	3	American Pharmacist Association

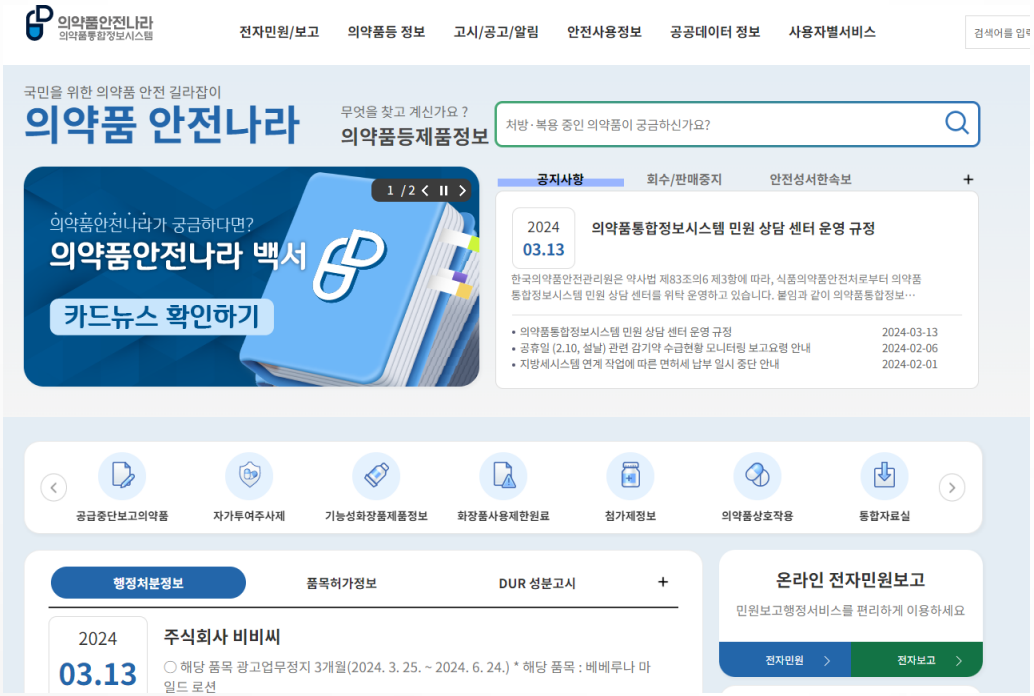
[그림 출처] 임상실무약학, 2024

5.3 EBM/P resources

- **국내 의약정보원**

  a. 식품의약품안전처 의약품안전나라

  - https://nedrug.mfds.go.kr/
  - 의약품 허가사항, 안전성 서한, 회수/폐기, 행정처분 등의 정보 확인 가능



5.3 EBM/P resources

- **국내 의약정보원**

  b. 약학정보원

  - **https://www.health.kr/**
  - **국내제조/수입의약품 정보 제공, 의약품 식별 관련 정보량이 많고 정확함**



5.3 EBM/P resources

- **국내 의약정보원**

  c. KIMS 의약품정보센터

  - https://www.kimsonline.co.kr/
  - 대체 가능 의약품 정보, 생산발매 현황 등의 확인이 용이함



5.3 EBM/P resources

- **국내 의약정보원**

  d. 드럭인포

  - https://www.druginfo.co.kr/

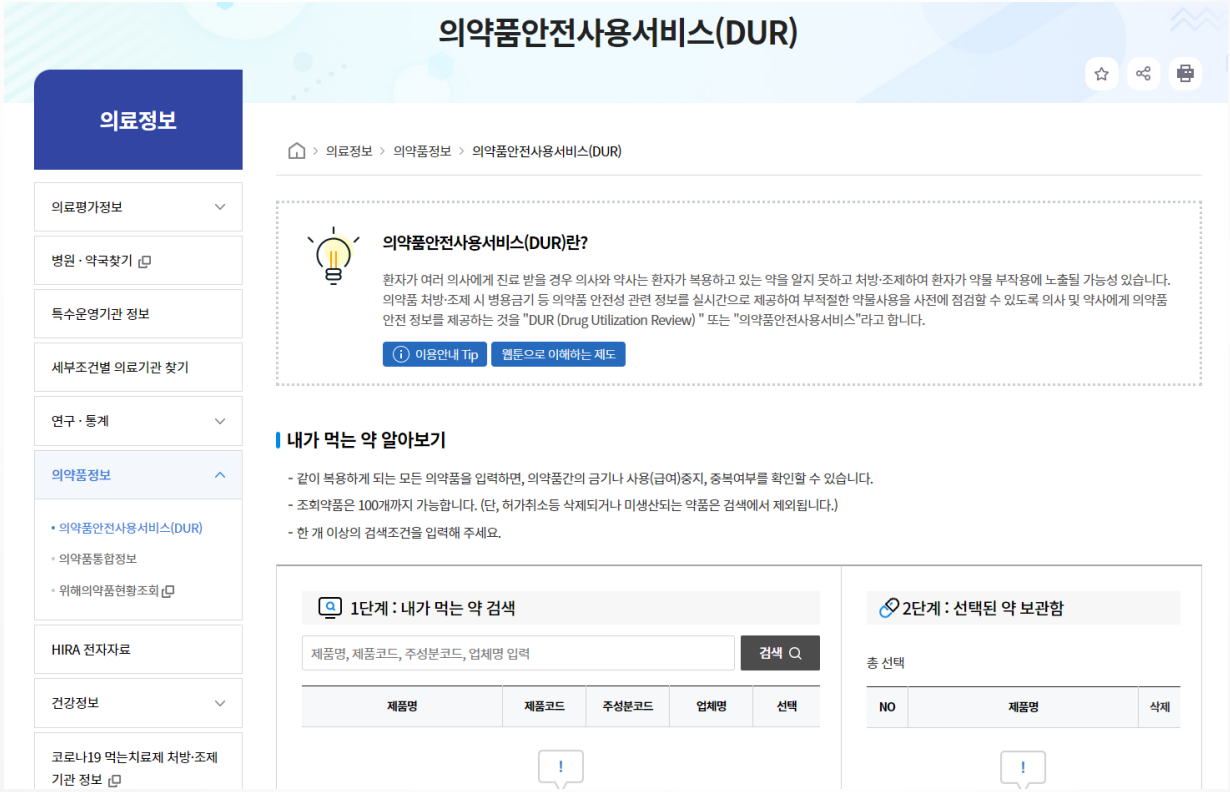


5.3 EBM/P resources

국내 의약정보원

e. 건강보험심사평가원

- https://www.hira.or.kr/
- 보험인정 기준, 약제 기준 정보, 의약품 통합정보 등



5.3 EBM/P resources

국내 의약정보원

f. 한국희귀필수의약품센터

- https://www.kodc.or.kr/



5.3 EBM/P resources

- **Summary of the primary evidence**
  - ACP Journal Club, Clinical Evidence, Dynamed, eMEdicine, Essential Evidence Plus, FPIN Clinical Inquiries, UpToDate

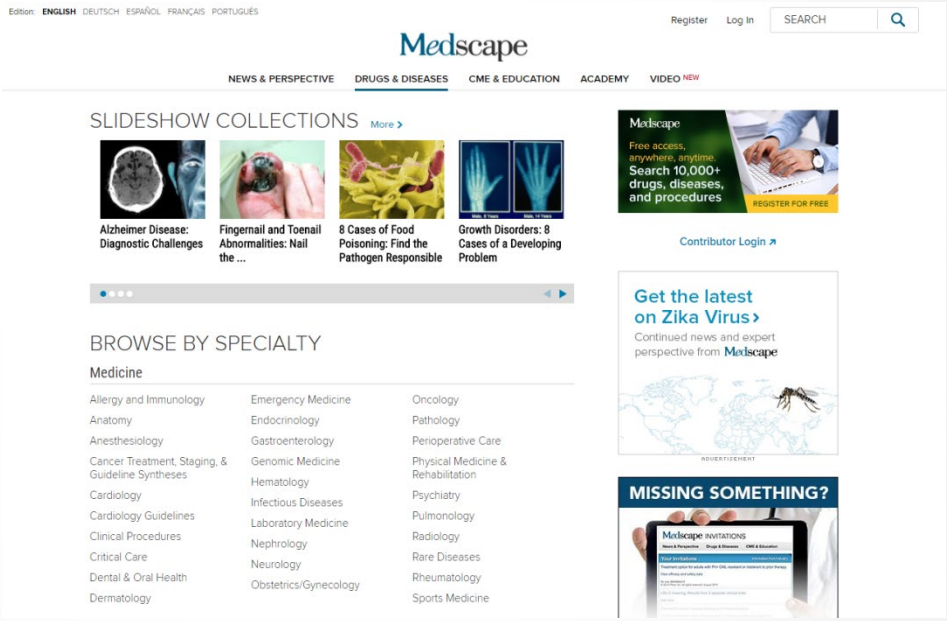
- **Electronic textbooks & libraries**
  - AccessMedicine, ACPMedicine, ACP PIER, Stat!Ref, First Consult, MD Consult

5.3 EBM/P resources

- **Summary of the primary evidence**

  a. eMedicine

  - https://emedicine.medscape.com/
  - A point-of-care medical reference available to health care professionals on the internet
  - Updated regularly by more than 8,000 physician or health care provider authors and editors

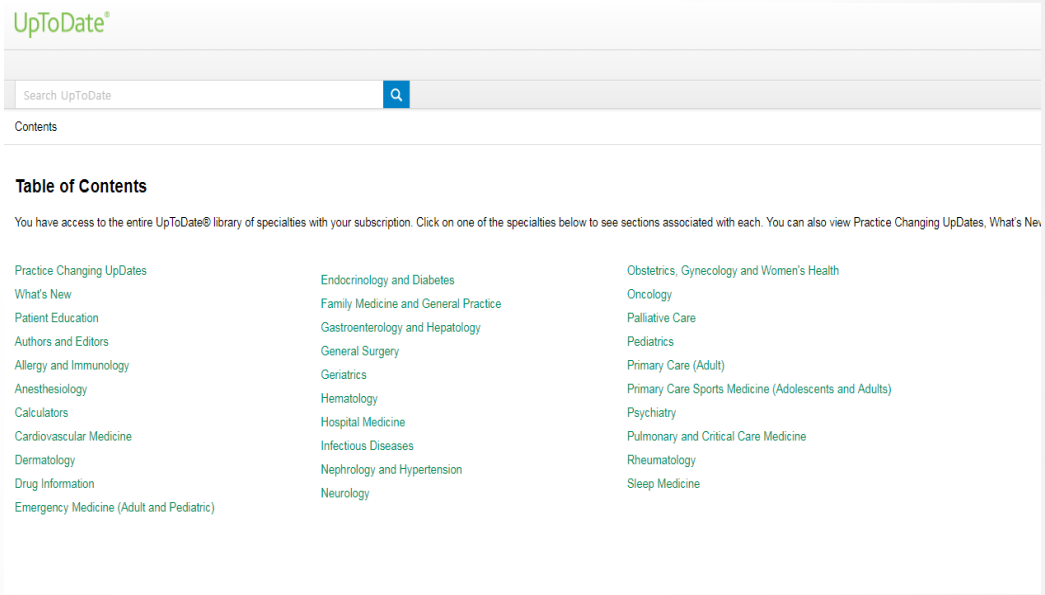


5.3 EBM/P resources

- **Summary of the primary evidence**

  b. UpToDate

  - http://www.uptodate.com
  - An evidence-based, peer reviewed information resource available via the Web, desktop/laptop computer, and PDA/mobile device

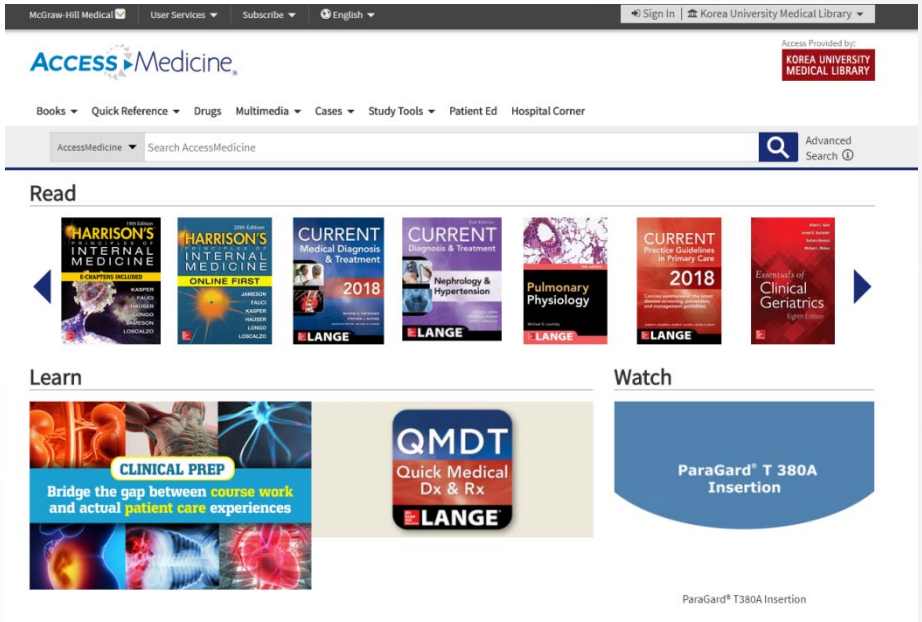


5.3 EBM/P resources

- **Electronic textbooks and libraries**

  a. AccessMedicine

  - http://accessmedicine.com/features.aspx
  - An online resource that provides students, residents, clinicians, researchers, and all health professionals with access to "Harrisons Online"



감사합니다!